

Physics of Welding

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Lesson Objectives

- Heat input and heat transfer from arc to weld
- Metal melting and regions of weld.
- Arc physics and plasma properties of arc

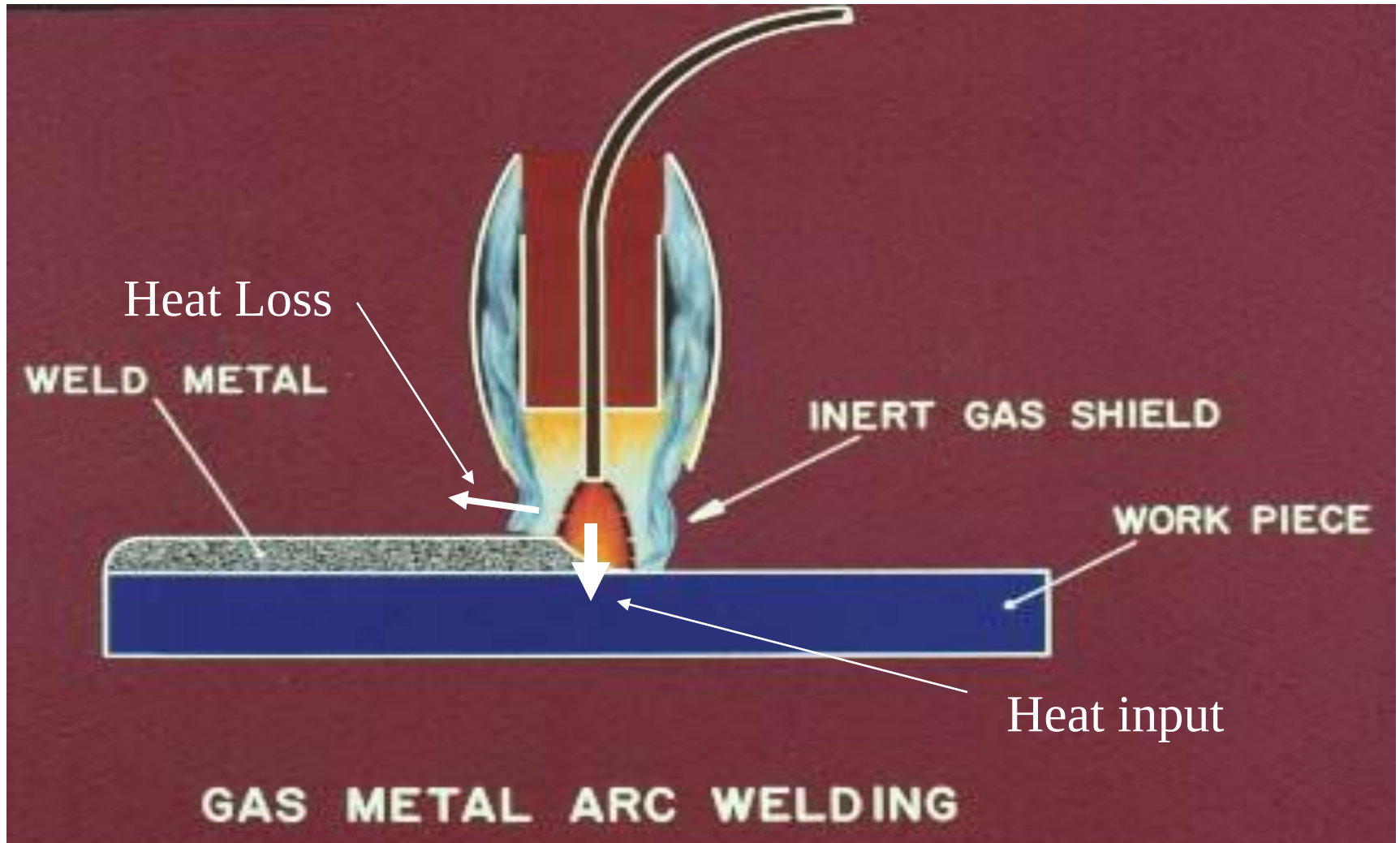
Keywords

Heat Input, Heat Transfer Efficiency, Heat Affected Zone, Enthalpy of Melting, Latent Heat, Melting efficiency, Plasma, Polarity, Thermionic Work Function, Ionization, Cathode Spot, Anode Spot, Arc I-V Characteristics

Physics of Welding

- **Heat Input Concepts**
- **Energy Sources**
- **Arc Characteristics**
- **Wire Melting**

Heat Input



Heat Input

H = energy input, energy/unit length, joules /mm

H = Power/Travel Speed, = P/v

P = total input power, Watts

v = travel speed of heat source, mm/sec

**Describes energy per unit length delivered,
not rate of delivery**

This energy does not all go entirely to the work

Heat Input for Arcs

$$H = P/v = EI/v$$

E = Arc Voltage (Volts)

I = Arc Current (Amps)

EI = Process power, converted to Heat

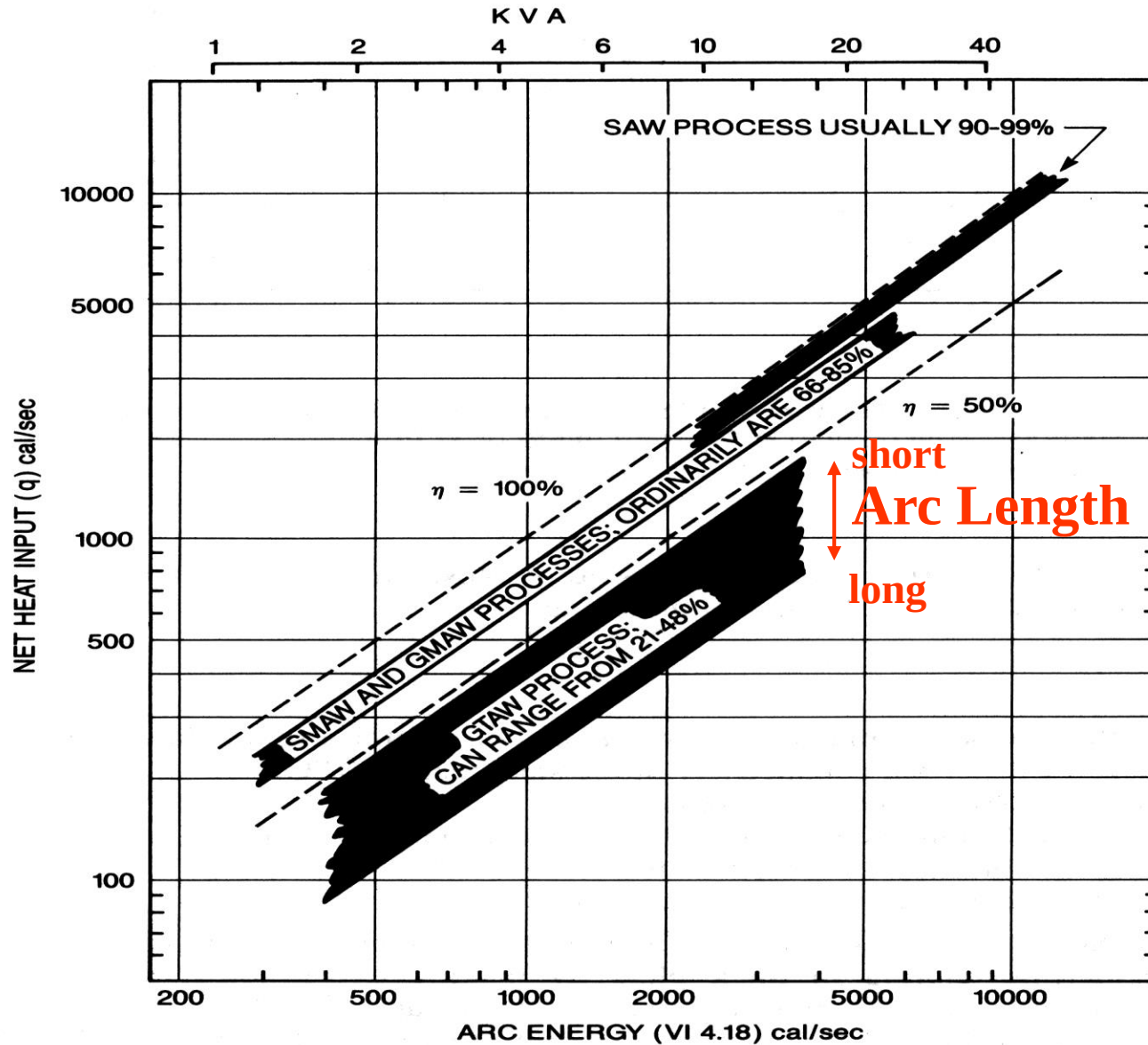
v = Welding Travel Speed

Not all the arc energy goes into the work

$$H_{\text{net}} = f_1 H = f_1 P/v = f_1 EI/v$$

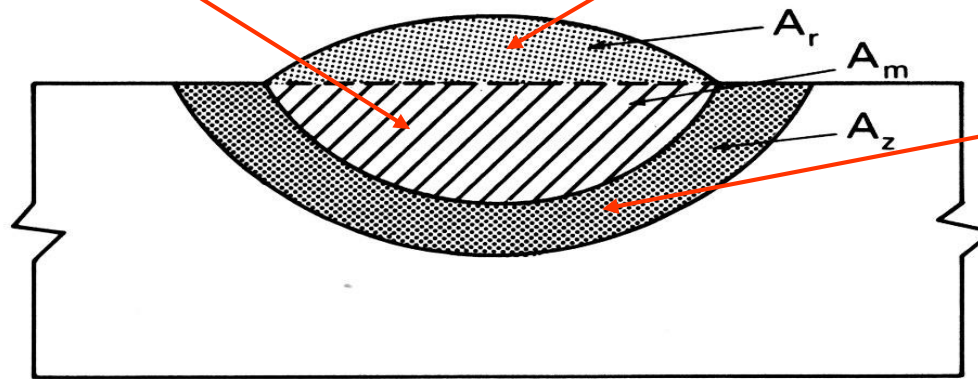
f_1 = Heat Transfer Efficiency

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Melted Base Metal

Reinforcement



**Heat
Affected
Zone**

$$A_w = \text{Cross Section of Weld} = A_m + A_r$$

For Autogenous Weld (no filler metal)

$$A_w = A_m$$

$$Q = \text{Heat Required to melt a Given Volume of Weld} = \text{Heat Required to elevate solid to MP} + \text{Latent Heat of Fusion}$$

Enthalpy of Melting

$$Q = \text{Heat Required to melt a Given Volume of Weld} = \text{Heat Required to elevate solid to MP} + \text{Latent Heat of Fusion}$$

$$Q = \rho C_p (T_m - T_o) + L$$

ρ = Density (mass / volume)

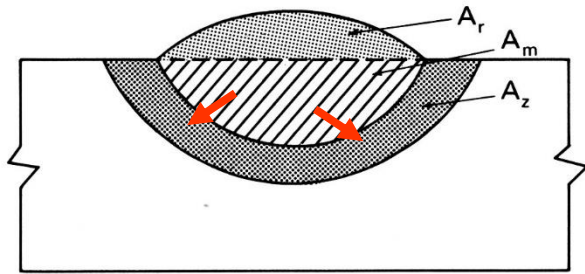
C_p = Heat Capacity (thermal energy / mass °C)

T_m = Melting Temperature

T_o = Initial Temperature, usually room temperature

L = Latent Heat of Fusion

Not all the net heat transferred goes into melting



Melting Efficiency

f_2 = Melting efficiency, the fraction of the process heat energy per unit length delivered to the metal which is required to melt the metal

$$f_2 = QA_w/H_{\text{net}}$$

$$f_2 = QA_w v / f_1 EI$$

From previous slide:

$$H_{\text{net}} = f_1 H = f_1 P / v = f_1 EI / v$$

Melting Efficiency Depends On:

- Higher Thermal Conductivity - Lower Efficiency
- High Energy Density Heat Source - Higher Efficiency

Other Energy Sources

Arc $H = EI/v$

Resistance: $H = I^2 R t$

Electroslag: $H = E I t$

Laser: $PD = \frac{4P_1}{\pi(f\theta)^2}$

EB: $PD = \frac{EI}{A}$

H = Heat generated, joules

E = Voltage, volts

v = Travel Speed, mm/sec

I = Current, amps

R = Resistance, ohms

t = Time, sec

PD = Power Density

P₁ = Input power

f = Focal length of lens

$\theta \cong \frac{\lambda}{\alpha}$ = angle of beam divergence

λ = wavelength

α = laser constant

A = Area of focused beam

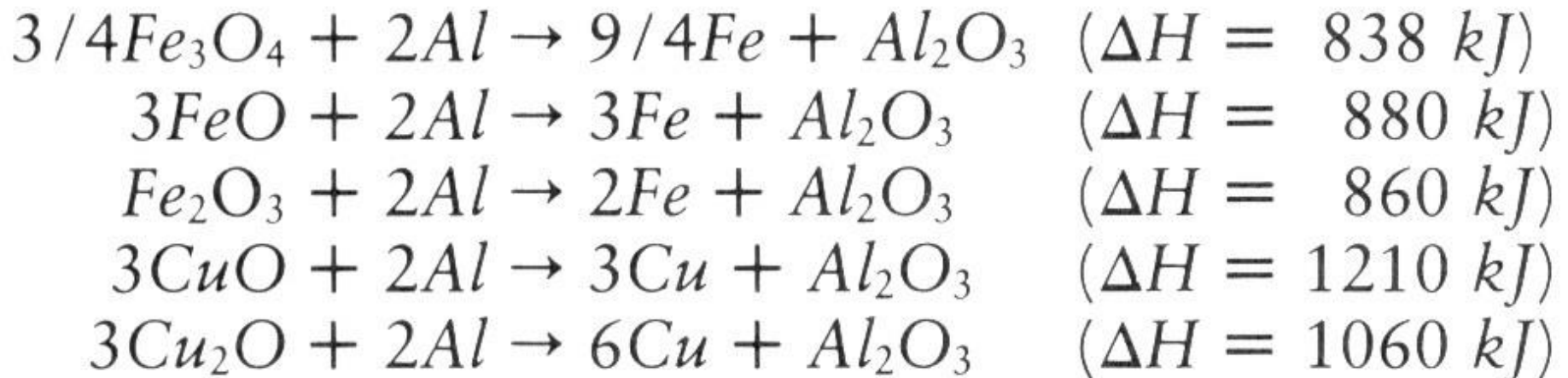
OXYFUEL GAS WELDING

Table 2.1
Flame Temperatures of Oxyfuel Gases

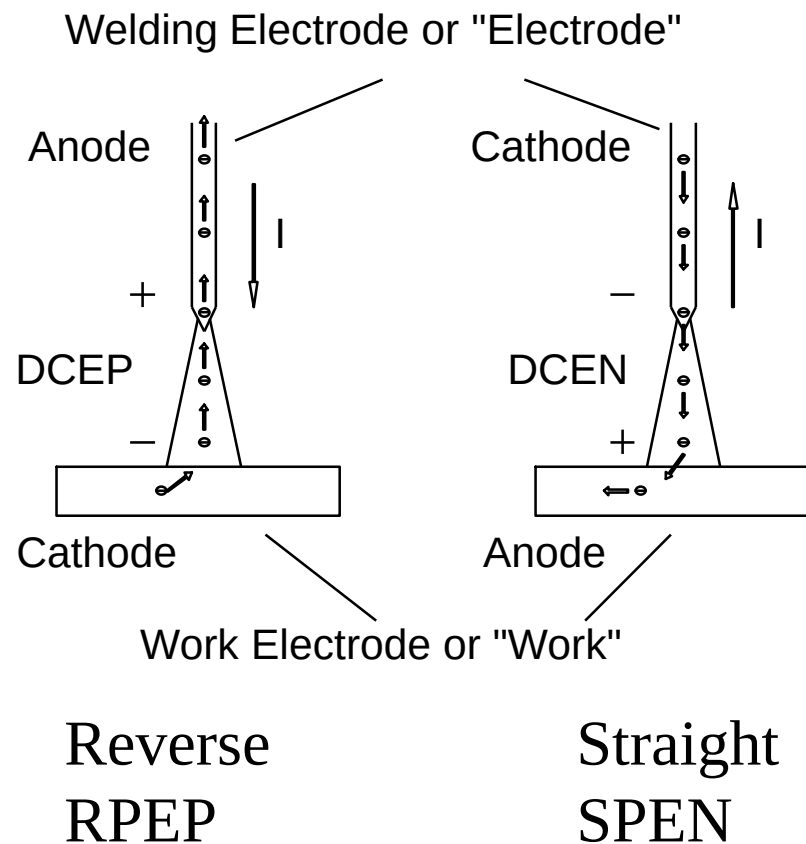
	Max Temp.		Neutral Flame Temp.	
	°C	°F	°C	°F
Acetylene	3100	5615	3100	5600
MAPP†	2900	5255	2600	5100
Propylene	2860	5174	2500	4900
Hydrogen	2870	5260	2390	4800
Propane	2780	5030	2450	4500
Natural gas/methane	2740	4967	2350	4260

†Methylacetylene-propadiene (stabilized)

THERMIT WELDING



Polarity and Current Flow



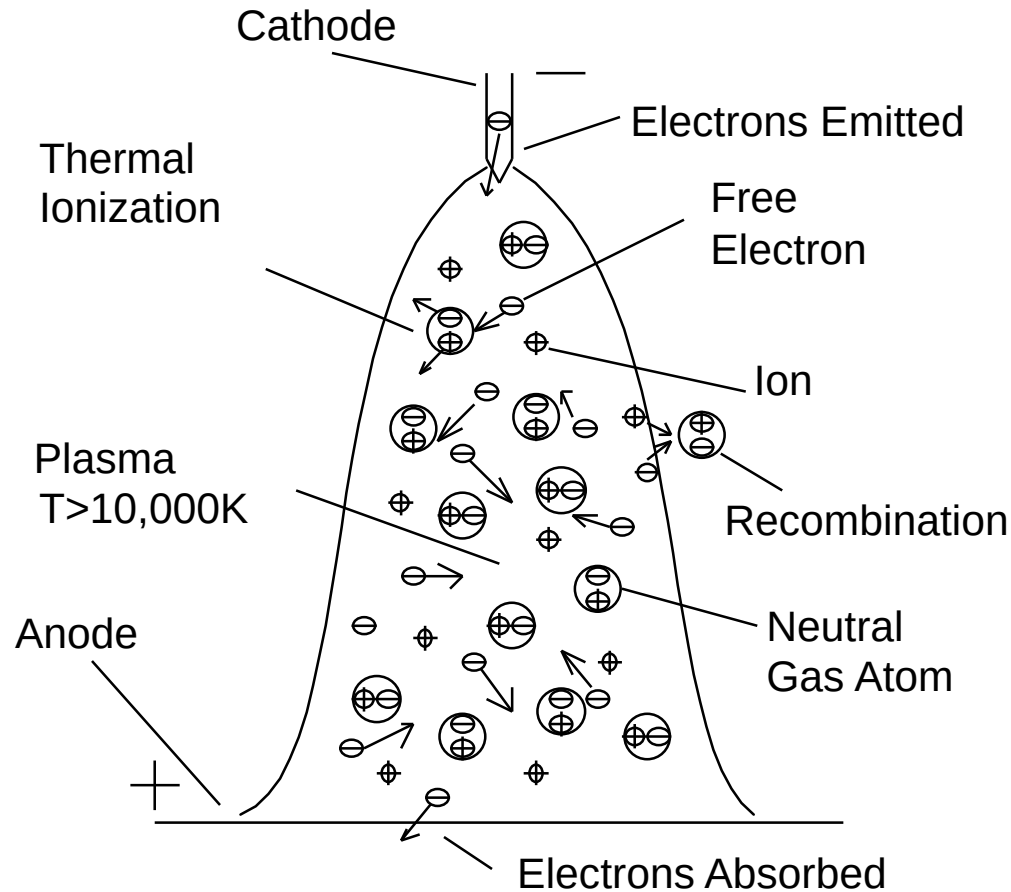
Plasma State

Gas is hot enough so that high energy collisions produce free electrons



Plasma may only be a few % electrons

Conduction of Current in the Arc



$$\frac{n_e n_i}{n_o} = \frac{2 Z_i (2\pi m_e k T)^{3/2}}{Z_o h^3} \exp \left\{ \frac{-V_i}{kT} \right\} \quad (2.18)$$

where

n_e , n_i , and n_o = particle densities (number per unit volume for electrons, ions, and neutral atoms, respectively)

V_i = the ionization potential of the neutral atom

T = temperature in degrees absolute

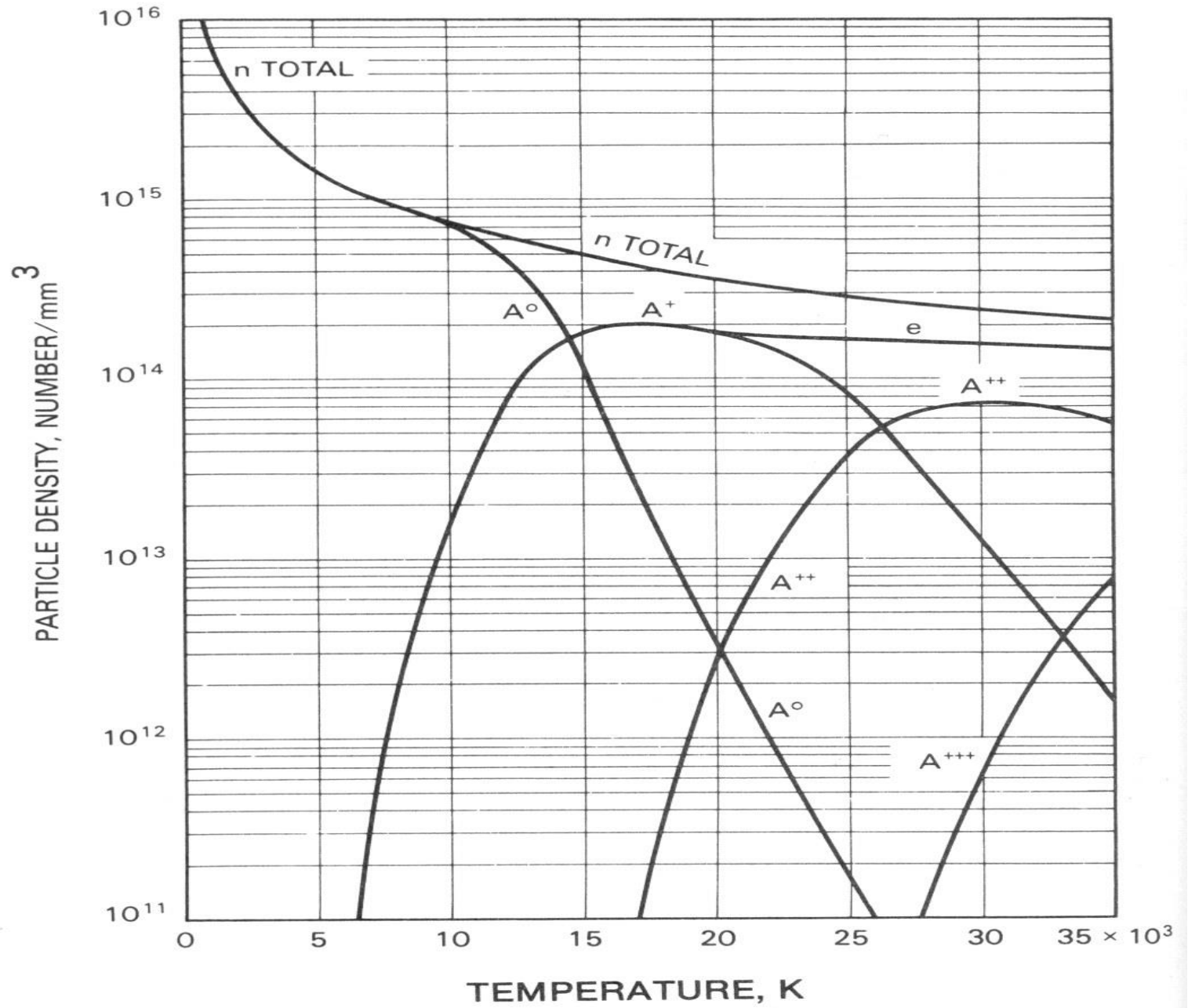
Z_i and Z_o = partition functions for ions and neutral particles

h = Planck's constant

m_e = the electron mass

k = Boltzmann's constant

Argon Arc



**APPROXIMATE
TEMPERATURES
ON APPROACH
TO ELECTRODE TIP
AND INTO PLASMA**

°C	°F
2425	4400
2700	4900
2760	5000
2930 (TIP)	5300
6075	11 000
(PLASMA)	

**TUNGSTEN
ELECTRODE**

**CATHODE
(-)**

**AREA OF SECONDARY
ELECTRON EMISSION**

**CATHODE SPOT AREA OF
PRIMARY ELECTRON EMISSION**

AREA OF CATHODE POTENTIAL DROP

**GENERAL DIRECTION OF
POSITIVE ION TRAVEL**

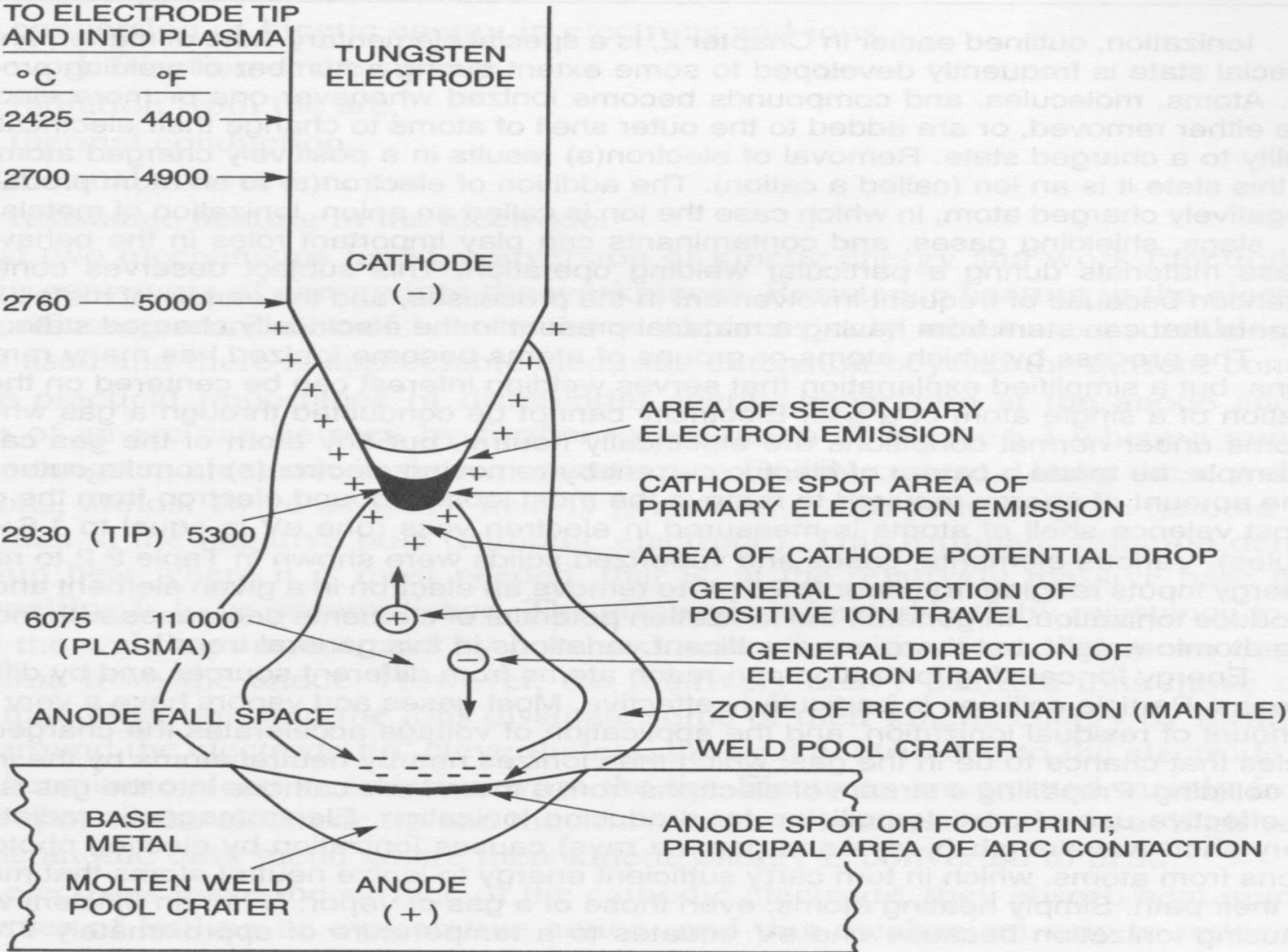
**GENERAL DIRECTION OF
ELECTRON TRAVEL**

ZONE OF RECOMBINATION (MANTLE)

WELD POOL CRATER

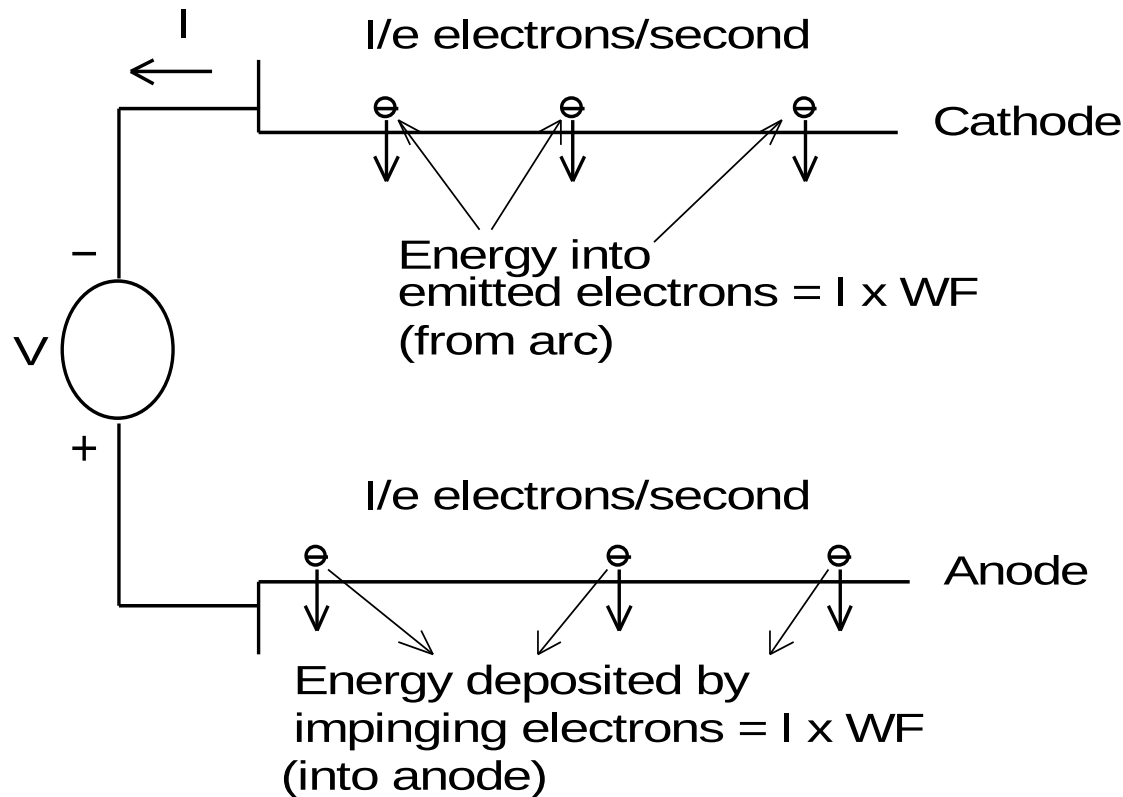
**ANODE SPOT OR FOOTPRINT;
PRINCIPAL AREA OF ARC CONTACTION**

**ANODE
(+)**



Thermionic Work Function

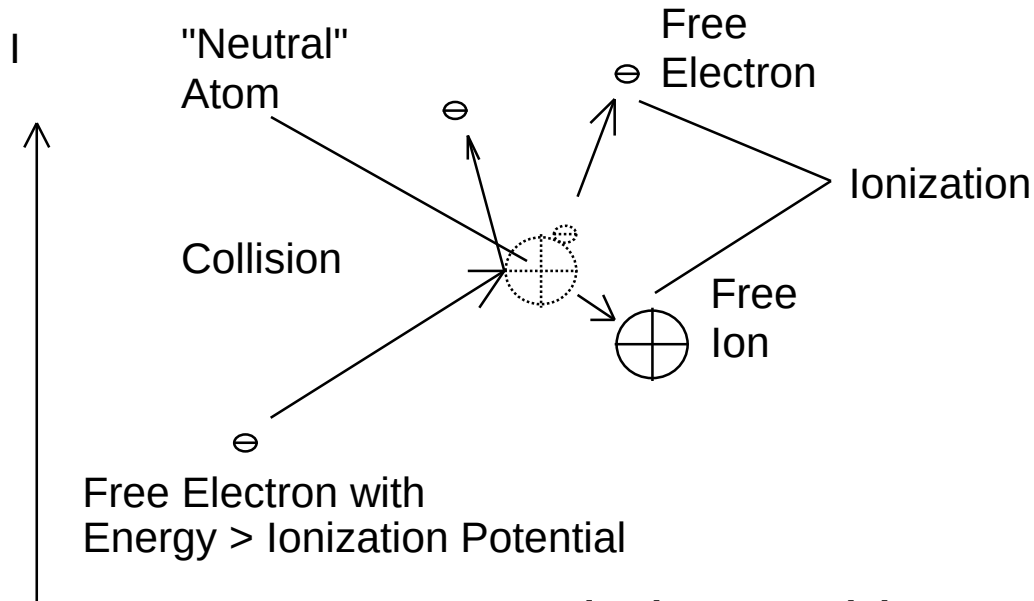
Energy Required for electron to escape a solid surface



Work Function of pure Tungsten = 4.4 eV

Work Function of Thoriated W = 4.1 eV

Ionization



Ionization Potentials:

He	24.6 eV
Ar	15.8
N	15.6
Fe	7.9
Na	5.1

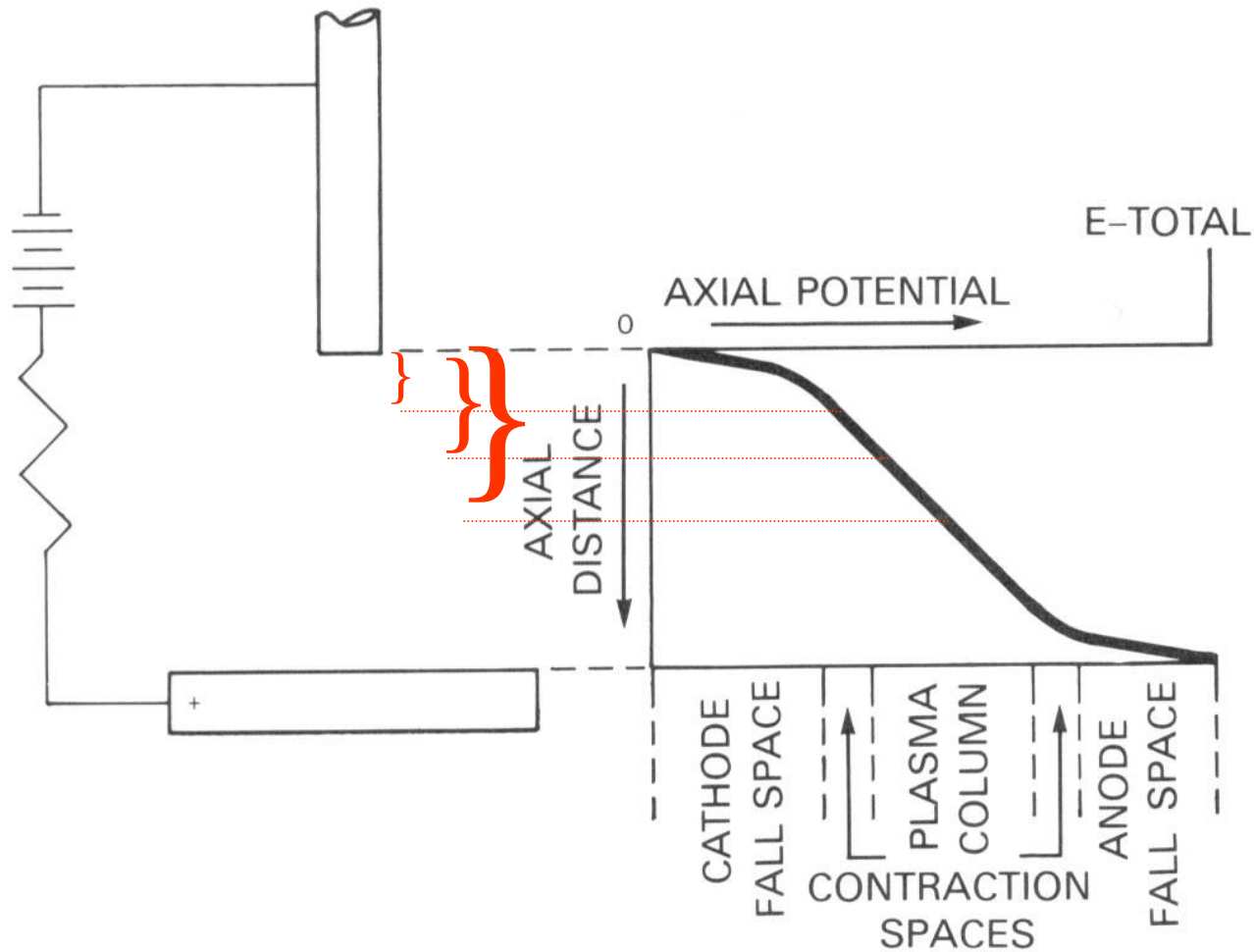
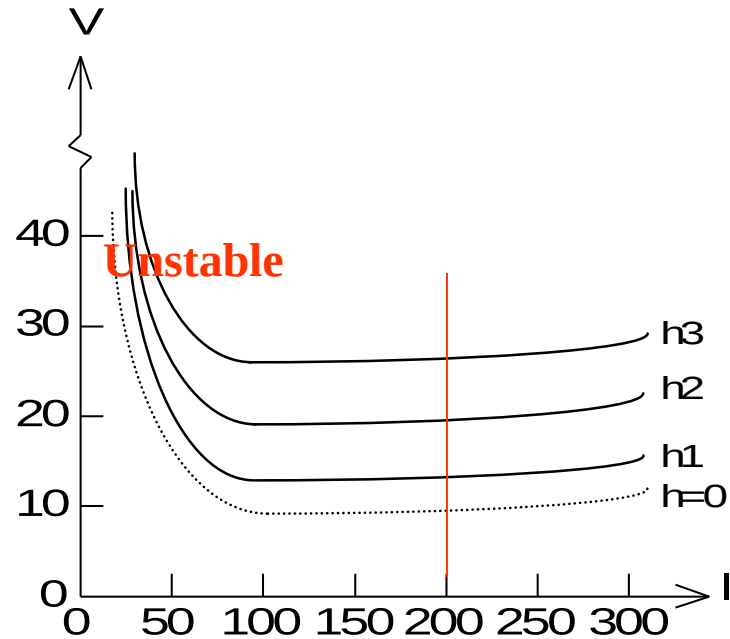
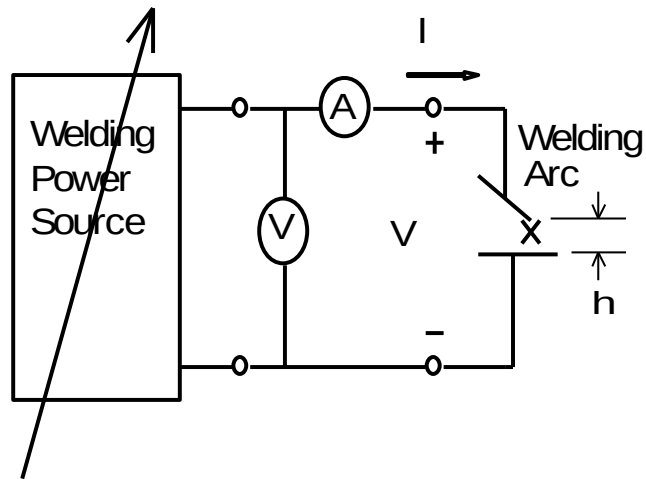


Figure 2.6—Arc Potential (Volts) Distribution Between Electrode and Work

Will total voltage change if we change the amount of current (say from 200 amps to 300 amps)?

Arc V-I Characteristic



Ionization Potential

He 24.6 eV

Ar 15.8

N 15.6

Fe 7.9

Na 5.1

P 4.3

